MODULE 16 / LESSON 01

RAG Quality and LLM Evaluation

Measure retrieval and generation separately so a failed answer can be diagnosed.

DSACADEMY

WHY THIS MATTERS

Start from what you already know

RAG ingestion, embeddings, retrieval, citations, and classification metrics

01 Explain retrieval precision and recall in plain language.

02 Use hybrid search and reranking in a small worked example.

03 Complete the guided lab and verify evaluation datasets and judges.

CORE CONCEPT

The vocabulary of this topic

Measure retrieval and generation separately so a failed answer can be diagnosed.

retrieval precision and recall: Retrieval precision measures result relevance; retrieval recall measures coverage of all relevant evidence.

hybrid search and reranking: Hybrid search combines lexical and semantic candidates; reranking applies a stronger model to reorder them.

groundedness: Groundedness measures whether generated claims are supported by the context supplied to the model.

evaluation datasets and judges: Evaluation datasets represent real tasks; judges apply explicit rubrics that must be checked against humans.

FOUR CONNECTED IDEAS

How the concept works

01

retrieval precision and recall →

02

hybrid search and reranking →

03

groundedness →

04

evaluation datasets and judges

Recall@k measures relevant evidence found; precision@k measures retrieved evidence that is relevant; generation metrics judge use of evidence.

CORE CONCEPT

Read every part of the mechanism

Recall@k measures relevant evidence found; precision@k measures retrieved evidence that is relevant; generation metrics judge use of evidence.

top-k: The number of retrieved candidates.

reranker: A second model that reorders candidates.

groundedness: Whether claims are supported by supplied context.

WORKED EXAMPLE

Worked example by hand

At k=4, retrieving three relevant chunks when five exist gives precision 3/4 and recall 3/5.

01

Known values

02

Apply the mechanism

03

Interpret the result

RUN IT AND INSPECT THE RESULT

Map the concept to Python

```
relevant_retrieved, retrieved, relevant_total = 3, 4, 5
print(relevant_retrieved / retrieved, relevant_retrieved / relevant_total)
```

OUTPUT

```
0.75 0.6
```

Each line corresponds to a concept already explained.

QUALITY GATE

Build the complete mental model

- ✓ Evaluate ingestion, retrieval, context assembly, and generation as separate failure boundaries.
- ✓ Hybrid retrieval combines exact-term and semantic strengths without directly adding unlike scores.
- ✓ Rerank only a small candidate set to improve quality while controlling latency and cost.
- ✓ Calibrate automated judges against human ratings and inspect disagreements rather than hiding them.

FOUR CONNECTED IDEAS

Notebook readiness map

01

retrieval precision and recall →

02

hybrid search and reranking →

03

groundedness →

04

evaluation datasets and judges

Use the concepts from this lesson to read and complete https://github.com/curiousily/AI-Bootcamp/blob/master/07.advanced-rag-with-llama-3-in-langchain.ipynb.

RISK REVIEW

Common beginner mistakes

01 **Changing retrieval and generation simultaneously, hiding which stage improved.**

02 **Trusting an LLM judge without calibrating it against human decisions.**

GUIDED LAB

Compare vector, keyword, hybrid, and reranked retrieval on a labelled question set and evaluate cited answers.

01

Predict

02

Run

03

Inspect

04

Explain

SESSION SYNTHESIS

Keep the system, not isolated definitions.

01 **Explain retrieval precision and recall in plain language.**

02 **Use hybrid search and reranking in a small worked example.**

03 **Complete the guided lab and verify evaluation datasets and judges.**

NEXT EVIDENCE

An evaluation report separating ingestion, retrieval, context, generation, latency, and cost failures.